

GRANT NEILSON

Computational Biologist with seven years of industry experience building scalable and robust analysis pipelines, developing internal Python & R packages and processing large-scale whole genome and single cell sequencing datasets (1TB+). Committed to writing production-quality code through rigorous testing, version control, and thorough documentation. Active open-source contributor, passionate about developing reliable scientific software and advancing reproducible research through community-driven collaboration.



INDUSTRY EXPERIENCE

2026
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2023

Senior Computational Biologist

CoSyne Therapeutics

📍 London, UK

- Led analysis of large-scale genomic datasets to support biomarker discovery and target identification, translating complex results into clear, decision-ready outputs for cross-functional teams. Culminated in the successful partnership with a publicly traded biotech.
- Co-Developed and maintained internal & open source R and Python packages. For statistical modelling, data processing and visualisation. Bixverse, is a Rust based R package enabling the analysis of 3 million single cells on a standard laptop. Cosyne-cnutils is a python package used for the analysis of genomics data.
- Developed internal container system using Docker and ECR, to enable reproducible research. This system used CI/CD to build images robustly and securely.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.
- Co-developed an internal MCP server exposing CoSyne's omics lakehouse and public biomedical APIs to LLM agents. Authored the DuckLake SQL catalog and query layer: a curated, allowlisted table catalog encoding row grain, join keys and data caveats, with server-side SELECT-only validation and row caps forming the safety boundary for model-generated SQL. Connected to Claude, enabling agents to reason across internal and public data and generate testable hypotheses.
- Led ingestion for the company DuckLake lakehouse, a Postgres catalog over Parquet on S3 queried with DuckDB, landing 70+ schema-validated tables spanning internal RNA-seq, WGS, methylation, CRISPR and dose-response data alongside OpenTargets, DepMap, CCLE/PRISM and Orphanet. Built a polars and pandera validation layer enforcing a declared schema on every write, shipped via Docker/ECR images and GitLab and GitHub CI.

View this CV [online](#)

CONTACT

✉ grant.neilson5@gmail.com

🐙 [GitHub](#)

in [LinkedIn](#)

LANGUAGES

📊 R
🐍 Python
🔗 Bash
🔗 Nextflow
🗄️ SQL

TECHNOLOGIES

🔗 GitLab/GitHub
🐳 Docker
📦 AWS
🖥️ Azure
🗄️ DuckDB
🔗 MCP/LLM Agents

OPEN SOURCE & PUBLIC PROJECTS

📊 [bixverse](#)
📊 [manifoldsR](#)
🔗 [cosyne-cnutils](#)
🔗 [Sarek](#)

Made with the R package [pagedown](#).

The source code is available on [GitHub](#).

Last updated August 2026.

2023
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2020

Computational Biologist

Fios Genomics

📍 Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

2020
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2019

Computational Biologist

Complex Diseases Epigenetics group, Exeter Medical School

📍 Exeter, UK

- Developed novel bioinformatics pipelines for processing genomics data.
- Gained an understanding of regression analysis, how to account for covariates, feature selection and optimal model fitting.



EDUCATION

2019
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2015

MSci Biology

University of Southampton

📍 Southampton, UK

- Grade : First Class Honours
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015
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2013

School

Merchiston Castle School

📍 Edinburgh, UK

- A Levels: Biology, Chemistry, Maths



SELECTED PUBLICATIONS

2025

TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas

BioRxiv

- L. Laaniste, J. Luginbuehl, M. Trinh, R. Watkins, G. Lueg, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.11.20.688938>

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.04.17.649387>

2025

• **Detecting cell level transcriptomic changes of Perturb seq using
Contrastive Fine tuning of Single Cell Foundation Models**

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, G. Neilson, et al.
- doi: <https://doi.org/10.1101/2025.04.17.649395>